SMART WATER TANK

MONITORING & CONTROL SYSTEM

Hardware System Requirements Specification (SRS)

ESP32 Microcontroller | HC-SR04 Ultrasonic Sensor | Relay Module

Covers: Hardware, Wiring, Sensors, Power, Safety, Enclosure, Upgrades

Version 1.0 | 2026

1. Project Overview

The Smart Water Tank Monitoring and Control System is an IoT hardware project designed to automate the monitoring and control of domestic or commercial water tanks. The system uses an ESP32 microcontroller, an ultrasonic distance sensor, and a relay module to continuously measure water levels and control a motor pump without human intervention.

The web dashboard displays water level in real time, but all the intelligence — sensor reading, relay switching, dry run protection, scheduling — runs entirely on the ESP32 hardware with no dependence on any external server or internet connection.

1.1 Project Objectives

- **Continuously measure water level using non-contact ultrasonic sensing**
- **Automatically control the motor pump based on water level thresholds**
- **Protect the motor from dry-run damage**
- **Provide physical and remote manual override capability**
- **Log all motor events with timestamps for diagnostic purposes**
- **Operate reliably 24x7 without manual intervention**
- **Survive power cuts and resume correct operation automatically**

1.2 System Scope

This document covers all hardware aspects of the system including component selection, electrical specifications, wiring diagrams, power requirements, safety considerations, physical enclosure, calibration, and future hardware upgrade paths. Software and web interface are excluded from this document.

2. Hardware Components

The following components form the complete hardware assembly of the Smart Water Tank system.

2.1 Core Components

Component	Model / Spec	Qty	Purpose
ESP32 Dev Board	ESP32-WROOM-32 / 38-pin	1	Main microcontroller - runs all logic and WiFi
Ultrasonic Sensor	HC-SR04	1	Measures distance from sensor to water surface
Relay Module	5V Single Channel Active LOW	1	Switches motor pump ON and OFF

Push Button	Momentary NO, 6mm or panel mount	1	Manual motor toggle
Ceramic Capacitor	100nF (0.1uF)	1	Button debounce filter
Breadboard	830 tie-point or PCB	1	Component interconnection
Jumper Wires	Male-Male and Male-Female	20+	Electrical connections
USB Cable	Micro-USB or USB-C per board	1	Programming and power
Power Supply	5V 1A minimum (phone charger)	1	ESP32 power after deployment

2.2 Optional But Recommended Components

Component	Specification	Why Recommended
Waterproof Enclosure	IP65 rated ABS box, 150x100x70mm min	Protects electronics from moisture and dust
JSN-SR04T Sensor	Waterproof ultrasonic, same pinout as HC-SR04	For outdoor tanks exposed to rain or humidity
10k Ohm Resistor	1/4 watt, carbon film	External pull-down for button in noisy environments
LED Indicator	5mm Red and Green	Visual motor status indicator on enclosure
220 Ohm Resistor	For LED current limiting	Protects LED from excess current
Terminal Block	2-pin screw terminal, 5mm pitch	Secure motor cable connections
DIN Rail Mount	35mm standard	Professional panel installation
Cable Glands	PG7 size, IP68	Waterproof cable entry into enclosure

3. Component Technical Specifications

3.1 ESP32 Microcontroller

Parameter	Value
Processor	Xtensa dual-core 32-bit LX6, up to 240 MHz
Flash Memory	4 MB (minimum for this project)
SRAM	520 KB
WiFi	802.11 b/g/n, 2.4 GHz only (not 5 GHz)

Bluetooth	BT 4.2 + BLE (not used in this project)
Operating Voltage	3.3V (GPIO logic level)
Input Voltage	5V via USB or VIN pin
GPIO Pins	34 programmable I/O pins
ADC	12-bit SAR ADC, up to 18 channels
Operating Temperature	-40 to +85 degrees C
Current Consumption	Active WiFi: ~160-260mA, Sleep: ~10uA

IMPORTANT: ESP32 GPIO pins operate at 3.3V logic. Never connect 5V signals directly to any GPIO pin. The HC-SR04 ECHO pin outputs 5V — a voltage divider is strongly recommended for reliable long-term operation.

3.2 HC-SR04 Ultrasonic Sensor

Parameter	Value
Operating Voltage	5V DC
Operating Current	15mA
Frequency	40 kHz ultrasonic
Minimum Range	2 cm
Maximum Range	400 cm (practical: 200 cm reliable)
Measuring Angle	15 degrees cone
Trigger Pulse	10 microsecond HIGH pulse on TRIG pin
Echo Output	HIGH pulse proportional to distance (5V level)
Resolution	0.3 cm
Accuracy	+/- 3 mm under ideal conditions

Distance Calculation Formula

The ESP32 measures the time the ECHO pin stays HIGH and calculates distance as:

```
Distance (cm) = Duration (microseconds) x 0.034 / 2
```

The division by 2 accounts for the sound wave travelling to the surface and back. The factor 0.034 is the speed of sound in cm per microsecond at room temperature.

Water Level Percentage Formula

```
Level (%) = ((emptyDist - distance) / (emptyDist - fullDist))
x 100
```

Where emptyDist is the distance from sensor to tank bottom (empty), and fullDist is the distance to water surface when tank is completely full.

Noise Filtering

The code takes 5 readings in rapid succession and averages only the valid ones (within expected range). Any reading outside the 1 to 40 cm window is discarded. If all 5 readings are invalid, the last known good reading is returned to prevent sudden false drops or spikes on the dashboard.

3.3 Relay Module

Parameter	Value
Coil Voltage	5V DC
Coil Current	70-80mA
Control Signal	Active LOW (motor ON when GPIO sends LOW)
Contact Rating (AC)	10A at 250V AC
Contact Rating (DC)	10A at 30V DC
Contact Type	SPDT (NO, NC, COM terminals)
Isolation	Optocoupler isolated input from output
Switching Time	~10ms
LED Indicator	Built-in LED shows relay state

WARNING: The relay switches high-voltage AC power to the motor. The AC wiring (COM and NO terminals) must only be connected by a qualified electrician. Incorrect wiring can cause electric shock, fire, or death. The ESP32 side of the relay is completely isolated from the AC side.

4. Pin Connections and Wiring

4.1 Complete Pin Assignment Table

ESP32 Pin	GPIO No.	Connects To	Wire Colour	Notes
D14	GPIO 14	HC-SR04 TRIG	Yellow	Output — trigger pulse
D27	GPIO 27	HC-SR04 ECHO (via divider)	Green	Input — use voltage divider for 5V to 3.3V

D26	GPIO 26	Relay Module IN	Blue	Output — Active LOW
D25	GPIO 25	Push Button (one leg)	White	Input PULLUP — other leg to GND
3.3V	3.3V	HC-SR04 VCC	Red	NOT 5V — sensor works on 3.3V with reduced range
VIN / 5V	5V	Relay Module VCC	Red	Relay coil needs 5V to operate reliably
GND	GND	HC-SR04 GND	Black	Common ground
GND	GND	Relay Module GND	Black	Common ground
GND	GND	Push Button (other leg)	Black	Pulls button pin LOW when pressed

4.2 Voltage Divider for ECHO Pin (Recommended)

The HC-SR04 ECHO pin outputs 5V but the ESP32 GPIO maximum input is 3.3V. Running 5V into an ESP32 GPIO can damage the chip over time. A simple resistor voltage divider converts 5V to approximately 3.3V:

Component	Value	Connection
R1	1k Ohm	Between HC-SR04 ECHO pin and ESP32 GPIO 27
R2	2k Ohm	Between ESP32 GPIO 27 and GND

Output voltage = 5V x (2000 / (1000 + 2000)) = 3.33V — safe for ESP32 GPIO input.

> If you use the JSN-SR04T waterproof sensor instead, the ECHO pin outputs 3.3V directly and no voltage divider is needed.

4.3 Push Button Wiring

The push button uses the ESP32's internal pull-up resistor (INPUT_PULLUP mode). One leg connects to GPIO 25 and the other directly to GND. When pressed, it pulls the pin LOW which the code detects as a button press.

Place a 100nF ceramic capacitor across the button terminals (between GPIO 25 and GND) to filter electrical noise and prevent false triggering. Without this capacitor, the button may trigger randomly due to motor vibration or loose connections.

5. Sensor Installation and Placement

5.1 Mounting the HC-SR04 Inside the Tank

The ultrasonic sensor must be mounted at the very top of the tank, facing straight down toward the water surface. Correct mounting is critical for accurate readings.

Requirement	Specification	Reason
Mounting position	Top centre of tank lid or wall	Ensures sensor faces water surface directly
Minimum clearance from walls	5 cm on all sides	Prevents wall echo interference
Minimum distance from full water	2 cm	HC-SR04 minimum range is 2 cm
Maximum distance from empty bottom	400 cm (practical 200 cm)	Maximum sensor range
Sensor angle	Exactly vertical (90 degrees to water)	Angled mounting causes inaccurate readings
Cable entry	Sealed with waterproof gland or silicone	Prevents moisture entering enclosure

5.2 Common Sensor Placement Mistakes

- **Mounting near inlet pipe:** Water turbulence causes erratic readings. Keep sensor at least 30 cm away from water inlet.
- **Mounting at an angle:** Even 10 degrees off-vertical causes the sound wave to reflect sideways and miss the sensor.
- **Mounting too close to full water:** When water reaches within 2 cm of the sensor, readings become invalid. Always leave at least 3-4 cm gap.
- **Mounting near a wall corner:** Multiple reflections cause false readings. Always mount at the centre of the top surface.
- **Using HC-SR04 outdoors:** The standard HC-SR04 is not waterproof. Even humidity can damage it. Use JSN-SR04T for outdoor tanks.

5.3 Calibration Measurements Required

After mounting the sensor, measure and record these two values before entering them into the dashboard settings:

Measurement	How to Measure	Example Value
Empty Distance	Measure with ruler from sensor face down to tank bottom. Do this when tank is completely empty.	15.0 cm
Full Distance	Fill tank completely. Measure from sensor face down to water surface.	2.5 cm

Tank Capacity	From manufacturer label or calculate: length x width x height in litres	500 L

6. Power System

6.1 Power Requirements

Component	Voltage	Current Draw	Notes
ESP32 (active WiFi)	5V via USB / 3.3V internal	160-260 mA	Peak during WiFi transmission
HC-SR04 sensor	3.3V (from ESP32)	15 mA	Continuous during operation
Relay coil	5V (from ESP32 VIN)	70-80 mA	Only when motor is ON
Total system	5V	350-400 mA typical	Use 5V 1A supply minimum

6.2 Recommended Power Supply

- **Minimum rating:** 5V 1A (1000mA)
- **Recommended rating:** 5V 2A for headroom and reliability
- **Connector:** Micro-USB or USB-C depending on ESP32 board
- **Quality:** Use a branded charger — cheap chargers cause voltage fluctuations that reset the ESP32
- **Cable length:** Keep USB cable under 1 metre to avoid voltage drop

6.3 Power Cut Behaviour

When power is cut and restored, the ESP32 boots fresh and loads all settings from flash memory (Preferences library). The motor starts in the OFF state regardless of what it was doing before the power cut. This is a safety feature — the motor will only restart if Auto mode is active and the level is below threshold, or if the user manually turns it on.

6.4 Battery Backup (Optional)

To keep the system running during power cuts, a UPS or battery backup can be added:

- **Option A:** 18650 LiPo battery + TP4056 charging module + MT3608 boost converter (5V output). Cheap and compact.
- **Option B:** Small UPS module (5V output, 2000mAh+). Plug-and-play, just connects in line with the power supply.

- **Option C:** 12V sealed lead acid battery + DC-DC converter. Higher capacity, suitable for 8+ hours backup.

With a 2000mAh 5V battery, the system runs approximately 5-6 hours without mains power. The relay and motor are powered separately from mains AC so battery backup only keeps monitoring alive, not the motor itself.

7. Relay Module and Motor Connection

7.1 How the Relay Works

The relay is an electrically operated switch. When the ESP32 sends a LOW signal (0V) to the relay IN pin, the relay coil energises and the switch closes, completing the motor circuit. When the ESP32 sends HIGH (3.3V), the relay opens and the motor stops.

The relay has three terminals on the AC side:

- **COM (Common):** Always connected. One mains wire connects here.
- **NO (Normally Open):** Disconnected when relay is off. Motor wire connects here. Circuit closes when relay activates.
- **NC (Normally Closed):** Connected when relay is off. Not used in this project.

HIGH VOLTAGE WARNING: The COM and NO terminals carry 220V AC mains power. These connections must be made by a licensed electrician only. Touching these terminals when powered can cause electrocution. Always disconnect mains power before any wiring work.

7.2 Motor Compatibility

Motor Type	Compatible	Notes
Single phase AC submersible pump	Yes	Most common domestic tank pump. Direct relay connection.
Single phase AC centrifugal pump	Yes	Direct relay connection.
3-phase AC motor	No - needs contactor	Relay cannot handle 3-phase. Use relay to trigger a contactor.
12V DC pump	Yes	Connect relay between 12V supply and pump. Use appropriate relay rating.
24V DC pump	Yes	Same as 12V DC, ensure relay contact rating matches.
High current pump over 10A	No - needs contactor	Relay contacts rated 10A max. Use relay to trigger industrial contactor.

7.3 Contactor for High Power Motors

If your motor draws more than 10A or is 3-phase, use the relay to trigger an industrial contactor instead of connecting the motor directly:

1. ESP32 GPIO 26 controls the relay as normal
2. Relay COM and NO connect to the contactor coil (typically 230V AC coil)
3. Contactor main contacts handle the full motor power
4. This provides full isolation and can handle motors up to hundreds of amperes

8. Dry Run Protection

8.1 What is Dry Run

Dry run occurs when the motor pump runs without water flowing through it. This happens when the water source (underground sump, municipal supply, borewell) is empty or has an air lock. Running a pump dry for even a few minutes can burn out the motor seals, bearings, and windings, requiring expensive repair or replacement.

8.2 How the System Detects Dry Run

The system uses a water level rise check to detect dry run:

5. When the motor turns ON, the current water level is recorded.
6. After a configurable wait time (default 5 minutes, settable in seconds), the current level is checked again.
7. If the level has not risen by at least the minimum rise threshold (default 2%), the system assumes dry run.
8. The motor is immediately stopped and a DRY RUN DETECTED warning is shown on the dashboard.
9. The motor cannot be restarted by Auto or Schedule mode until manually cleared.

LIMITATION: Dry run detection works by watching the water level rise. If the sensor is misaligned or giving wrong readings, the system may falsely detect dry run even when water is flowing. Always verify sensor calibration is correct.

8.3 Hardware Dry Run Switch (Upgrade)

A more reliable dry run detection method uses a pressure switch or flow sensor:

- **Pressure switch:** Mounts on pump outlet. Opens circuit if water pressure drops below threshold. Completely independent of the ultrasonic sensor.
- **Flow sensor:** Measures actual water flow rate (YF-S201 water flow sensor, connects to GPIO). If flow rate is zero while motor is running, dry run is confirmed.
- **Float switch:** Mounted in the sump/source tank. Prevents motor from starting if source tank is empty.

9. Safety Systems

9.1 Software Safety Features

Safety Feature	How It Works
Emergency Stop	Single button on dashboard stops motor instantly and blocks Auto/Schedule from restarting it
Dry Run Protection	Auto-stops motor if water level does not rise after configured time
Auto OFF at full tank	Motor stops automatically when level reaches configurable threshold (default 90%)
Power cut safe state	Motor always starts in OFF state after power restoration
Button lock	Prevents physical button from interfering with web control for configurable seconds
Mode isolation	Manual mode completely disables all automatic control

9.2 Recommended Additional Hardware Safety

- **MCB (Miniature Circuit Breaker):** Install a 6A or 10A MCB in the motor circuit. Trips automatically on short circuit or overload. Protects relay, wiring, and motor.
- **ELCB / RCCB:** Earth Leakage Circuit Breaker protects against earth faults and water leakage on pump body. Essential for submersible pumps.
- **Surge protector:** Protects ESP32 from voltage spikes. Install on mains supply line.
- **Fuse on relay output:** A 10A fuse on the NO terminal protects relay contacts from overload.
- **Earthing:** Ensure motor body and water pipe are properly earthed. Critical for submersible pumps.

All mains electrical safety devices (MCB, ELCB, earthing) must be installed by a licensed electrician in compliance with local electrical codes. This is not optional — it is a legal and safety requirement.

10. Physical Enclosure

10.1 Enclosure Requirements

Requirement	Specification
Protection rating	IP65 minimum (dust tight, water jet resistant)

Material	ABS plastic or fibreglass — do not use steel (rusts near water)
Minimum internal dimensions	150mm x 100mm x 70mm for all components
Mounting	Wall mount with screws, or DIN rail mount inside electrical panel
Temperature rating	0 to 50 degrees C operating, store away from direct sun
Ventilation	None required — ESP32 heat output is minimal

10.2 Enclosure Layout

Recommended layout inside the enclosure from top to bottom:

10. Relay module at top — closest to cable entry for motor wires
11. ESP32 dev board in middle — central for all connections
12. Power input connector (USB or terminal block) at bottom
13. HC-SR04 sensor cable exits through IP68 cable gland on the side
14. Motor cable exits through separate IP68 cable gland on the other side
15. Push button mounts on enclosure lid with panel mount button

TIP: Use standoffs and M3 screws to mount the ESP32 and relay to the base plate inside the enclosure. Hot glue or double-sided tape loosens over time in humid environments.

10.3 Labelling

Label all cable glands, terminals, and connectors clearly:

- **SENSOR — cable to HC-SR04**
- **MOTOR LIVE / MOTOR NEUTRAL — AC motor connections on relay**
- **MAINS LIVE / MAINS NEUTRAL — incoming AC power**
- **5V POWER — DC power input for ESP32**
- **GPIO labels on any exposed test points**

11. Installation Checklist

Complete these steps in order before first power-on:

11.1 Mechanical Installation

16. Mount HC-SR04 sensor at top centre of tank, facing straight down
17. Verify minimum 2 cm clearance between sensor and maximum water level
18. Verify minimum 5 cm clearance from tank walls on all sides
19. Seal sensor cable entry with silicone or cable gland
20. Mount electronics enclosure on wall near tank, out of direct sun and rain

21. Run sensor cable from tank to enclosure through conduit if exposed

11.2 Electrical Installation (Licensed Electrician Required for AC Parts)

22. Install MCB and ELCB on mains supply to motor circuit
23. Connect motor live wire to relay NO terminal
24. Connect motor neutral directly to mains neutral (relay only switches live)
25. Connect motor earth to earth busbar
26. Connect mains live to relay COM terminal
27. Connect 5V power supply USB to ESP32
28. Verify all AC connections are tight and insulated with appropriate terminals

11.3 Electronics Wiring

29. Connect HC-SR04 VCC to ESP32 3.3V
30. Connect HC-SR04 GND to ESP32 GND
31. Connect HC-SR04 TRIG to GPIO 14
32. Connect HC-SR04 ECHO through voltage divider to GPIO 27
33. Connect Relay VCC to ESP32 VIN (5V)
34. Connect Relay GND to ESP32 GND
35. Connect Relay IN to GPIO 26
36. Connect push button between GPIO 25 and GND
37. Place 100nF capacitor across push button terminals

11.4 First Power-On Verification

38. Power on ESP32 via USB only (motor circuit disconnected)
39. Verify ESP32 connects to WiFi — check Serial Monitor
40. Open dashboard and verify water level reads correctly
41. Test relay by clicking ON on dashboard — LED on relay module should light up
42. Verify relay clicks and LED changes when pressing motor OFF
43. Test physical button — motor should toggle
44. Power off, connect motor circuit, power on
45. Test full motor operation — verify water starts flowing

12. Future Hardware Upgrades

The following hardware additions can significantly enhance the system beyond the current prototype stage.

12.1 Sensor Upgrades

Upgrade	Component	Benefit
Waterproof sensor	JSN-SR04T	Same pinout, fully sealed for outdoor or humid tanks
Higher accuracy sensor	RCWL-1601 or A02YYUW	Better accuracy, longer range, less noise
Water quality sensor	TDS meter module	Measures dissolved solids — detects contamination
Temperature sensor	DS18B20 waterproof	Water temperature monitoring
Flow meter	YF-S201 Hall effect	Measures actual litres per minute flow rate
Pressure sensor	MPX5010DP	Measures water pressure — better than ultrasonic in turbulent water

12.2 Control and Automation Upgrades

Upgrade	Component	Benefit
Source tank float switch	NC float switch	Prevents motor starting when source tank is empty
Motor current sensor	ACS712 30A module	Detects actual motor current — better dry run detection
Dual tank control	Second relay module	Control two tanks independently from one ESP32
Booster pump control	Second relay module	Auto-start booster pump when pressure drops
Solenoid valve	12V NC solenoid valve + relay	Auto-close water inlet when tank is full
Real time clock	DS3231 RTC module	Maintains accurate time without internet NTP

12.3 Display and Alerting Upgrades

Upgrade	Component	Benefit
LCD Display	16x2 I2C LCD	Shows level and motor status without opening dashboard
OLED Display	0.96 inch I2C OLED	Compact, high contrast display for key readings
LED Bar Graph	10-segment LED bar + driver IC	Visual tank level indicator on enclosure
Buzzer	5V active buzzer	Local alarm when tank is full or empty
GSM Module	SIM800L or A6 mini	SMS alerts when WiFi is unavailable

12.4 PCB and Permanent Installation

For a professional permanent installation, replace the breadboard prototype with a custom PCB:

- **Design a PCB using KiCad (free) or EasyEDA (online, free)**
- **Include all components on one board: ESP32, relay driver, voltage divider, button, LEDs, power regulation**
- **Order from JLCPCB or PCBWay (typically under $10 for 5 pieces)**
- **Add proper strain relief for all cables**
- **Consider conformal coating on the PCB for moisture protection**

13. Maintenance and Troubleshooting

13.1 Routine Maintenance Schedule

Task	Frequency	How
Clean sensor face	Monthly	Wipe HC-SR04 transducer faces with dry cloth — algae or mineral deposits reduce accuracy
Check all wire connections	Every 3 months	Pull test each connector — vibration can loosen terminals over time
Verify relay operation	Every 3 months	Listen for relay click when toggling from dashboard
Check power supply output	Every 6 months	Measure 5V at ESP32 VIN with multimeter
Inspect enclosure seals	Every 6 months	Look for cracks or yellowing in enclosure, replace gasket if needed
Calibration check	Every 6 months	Manually measure water level and compare with dashboard reading
Motor brush check	Annually	Per motor manufacturer's maintenance schedule

13.2 Common Hardware Problems and Solutions

Problem	Likely Cause	Solution
Level reading jumps randomly	Motor vibration causing ultrasonic interference OR loose sensor mount	Re-mount sensor firmly, increase averaging samples in code
Level always reads 0%	Sensor cable disconnected OR calibration wrong	Check TRIG/ECHO wiring, re-enter calibration values

Motor does not turn on	Relay not receiving signal OR relay faulty OR AC wiring issue	Check relay LED — if no LED, check GPIO 26 wiring. If LED on, check AC connections
Button triggers randomly	Missing 100nF capacitor OR long button wire acting as antenna	Add capacitor across button, use shielded cable for long button runs
ESP32 keeps restarting	Power supply too weak OR WiFi connection issue	Use 5V 2A supply, check WiFi signal strength
Relay clicks but motor does not run	AC wiring fault OR motor fault OR MCB tripped	Check MCB, verify AC connections, test motor independently
Sensor reading changes with temperature	Speed of sound varies with temperature	Normal behaviour — accuracy changes +/-1% per 5 degrees C

14. System Specifications Summary

Parameter	Specification
Microcontroller	ESP32-WROOM-32, dual core 240MHz
Operating voltage	5V DC input, 3.3V internal logic
Power consumption	350-400mA typical at 5V (1.75-2W)
WiFi standard	802.11 b/g/n, 2.4 GHz
Sensor type	HC-SR04 ultrasonic, non-contact
Sensor range	2 cm to 400 cm (practical 200 cm)
Level resolution	Approx. 0.3 cm distance = ~1-2% level accuracy
Sensor read interval	Every 5 seconds, 5-sample averaged
Motor control	5V active-LOW relay, 10A 250VAC contact rating
Operating temperature	0 to 50 degrees C
History buffer	Last 30 level readings in RAM
Settings storage	Flash memory (Preferences) — survives power cuts
mDNS hostname	watertank.local
OTA update	Wireless firmware upload over WiFi
Tank capacity supported	Any size — calibrated by empty/full distance measurement
Motor types supported	Single phase AC up to 10A, DC pumps up to 10A